

1948 US marine geologists Bruce Heezen and Marie Tharp start using sonar data to map the features of the ocean floors. They publish their World Ocean Floor map in 1977.

1954 Piloted by Georges Houot and Pierre Willm, the French research submersible *F.N.R.S. 3* dives to 13,255 ft (4,040 m) off the coast of Dakar, west Africa, pioneering a new era of manned submersibles.

1955 The US survey ship *Pioneer* tows the first marine magnetometer and discovers magnetic "stripes" on the ocean floor off the West Coast of the US.

1958 The American nuclear submarine *USS Nautilus* makes a voyage under the ice at the North Pole, proving that there is no land there.

1959–62 US geologist Harry Hess proposes his theory of ocean-floor spreading from midocean ridges.

1960 Jacques Piccard and Don Walsh descend into the deepest part of the Pacific—the Challenger Deep in the Mariana Trench—in the bathyscaphe *Trieste*.

1961 The US Scripps Institution of Oceanography starts work on the Deep Tow System, pioneering remotely operated oceanographic research.

1963 British scientists Frederick Vine and Drummond Matthews realize that magnetic stripes in the rocks of ocean floor provide proof of Hess's model of ocean-floor spreading.

1963 A volcano erupts from the Mid-Atlantic Ridge just south of Iceland, and creates the new island of Surtsey.

1963 Canadian geologist John Tuzo Wilson introduces the theory of hotspots, created in the Earth's crust by plumes of heat within the mantle.

1963 First operational multibeam sonar sounding system installed on the American survey ship *Compass Island*.

1964 The floor of the Pacific Ocean slides 65 ft (20 m) beneath Alaska, causing the most powerful earthquake ever recorded in North America.

1964 The submersible *Alvin* makes its first dive. Operated by the Woods Hole Oceanographic Institution in the US, it ushers in a new age of intensive research and exploration in the deep oceans.

1967 The La Rance tidal barrage begins operating near St. Malo, France, generating 240,000 kilowatts of electricity on each ebb tide.

1968 The drilling ship *Glomar Challenger* takes samples of rocks up to 0.6 miles (1 km) below the ocean floor, and confirms the theory of ocean-floor spreading.

1970 The worst tropical storm of the 20th century occurs in Bangladesh. High winds and flooding caused by the storm surge kill 300,000 to 500,000 people.

1977 Scientists diving to the ocean floor in *Alvin* discover hydrothermal vents near the East Pacific Rise.

1980 A single oil-well blowout from the *Ixtoc 1* drilling rig in the Bay of Campeche, Mexico, spills 525,000 tons (475,000 metric tons) of crude oil into the Gulf of Mexico.

1982–83 An unusually severe El Niño event disrupts the weather throughout the Pacific region. Stocks of fish off Ecuador and Peru in the Eastern Pacific are virtually destroyed.

1985 The dead zone in the Gulf of Mexico, caused by pollution carried into the sea by the Mississippi River, is systematically surveyed for the first time.

1985 A team led by Dr. Robert Ballard discovers the wreck of the *Titanic* on the floor of the North Atlantic, 13,000 ft (4,000 m) below the ocean surface.

1989 The giant oil tanker *Exxon Valdez* runs aground in Prince William Sound, Alaska, leaking 30 million gallons (114 million liters) of crude oil into the sea. The oil spill kills 250,000 seabirds and 6,000 sea otters and is the most destructive in history.

1990 US geophysicist Syukuro Manabe uses a computer model of world climate to show that global warming could shut down the Gulf Stream.

1995 The largest recorded ocean wave strikes the ocean liner *Queen Elizabeth II* off the coast of Newfoundland, Canada. The wave is 100 ft (30 m) high.

1995 Using radar data from the Geosat satellite, Walter Smith and Dave Sandwell make the first truly accurate map of the ocean floor.

1998 A major coral bleaching event caused by unusually high ocean water temperatures damages more than a quarter of the world's coral reefs.

1998 Surveyors searching for oil fields discover a huge cold-water coral reef in the North Atlantic, off Scotland. The reef lies at a depth of 3,300 ft (1,000 m) and covers 39 sq miles (100 sq km).

2000 The space probe *Galileo* provides evidence that Europa, one of Jupiter's moons, may have oceans of liquid water beneath its icy surface. If so, it would be the only other body in the solar system with oceans.

2002 The Larsen B ice shelf on the northwest side of the Weddell Sea in Antarctica collapses because of high water temperatures. The collapse is widely blamed on global climate change.

2003 The world's first open-sea tidal power generator is installed off the coast of north Devon in southern England. The system uses a turbine to harness the power of strong tidal streams.

2004 Measurements of ocean currents associated with the Gulf Stream show that the flow has slowed by 30 percent since the 1960s. They suggest that the Gulf Stream may be under serious threat.

2004 The Asian tsunami devastates coastal communities in the Indian Ocean. It causes more casualties than any other in recorded history, with more than 150,000 people killed and over 25,000 missing.

2005 A storm surge caused by Hurricane Katrina floods the city of New Orleans, Louisiana, causing the deaths of more than 1,000 people.

2005 An international team of oceanographers reports that the oceans are becoming dangerously acidified by carbon dioxide absorption from the air. This could have severe consequences for marine life.

2006 Scientists drilling into ocean-floor sediments off Canada discover frozen methane at much shallower depths than expected, raising the possibility that it could be exploited as a fuel, and the fear that it might contribute to climate change.

2006 A scientific study finds that fish stocks have collapsed in nearly one-third of sea fisheries, that the rate of decline is accelerating, and that if the trend continues, the oceans will be fished out by 2050.

2007 Scientists report that the Southern Ocean is soaking up less carbon dioxide than in the past, and that this is accelerating climate change.